

K-FIELD

INTERNAL ENERGY HIERARCHY

A foundational reference on electron realization, Earth-CMB phase energy, Cell3 node and transition energies, reciprocal rest-energy radials, Emily and A2 modes, full Lucas Group collective structure, and nuclear closure-lock pathways.

K-Field Internal Energy Hierarchy: Cell3 to Lucas Group

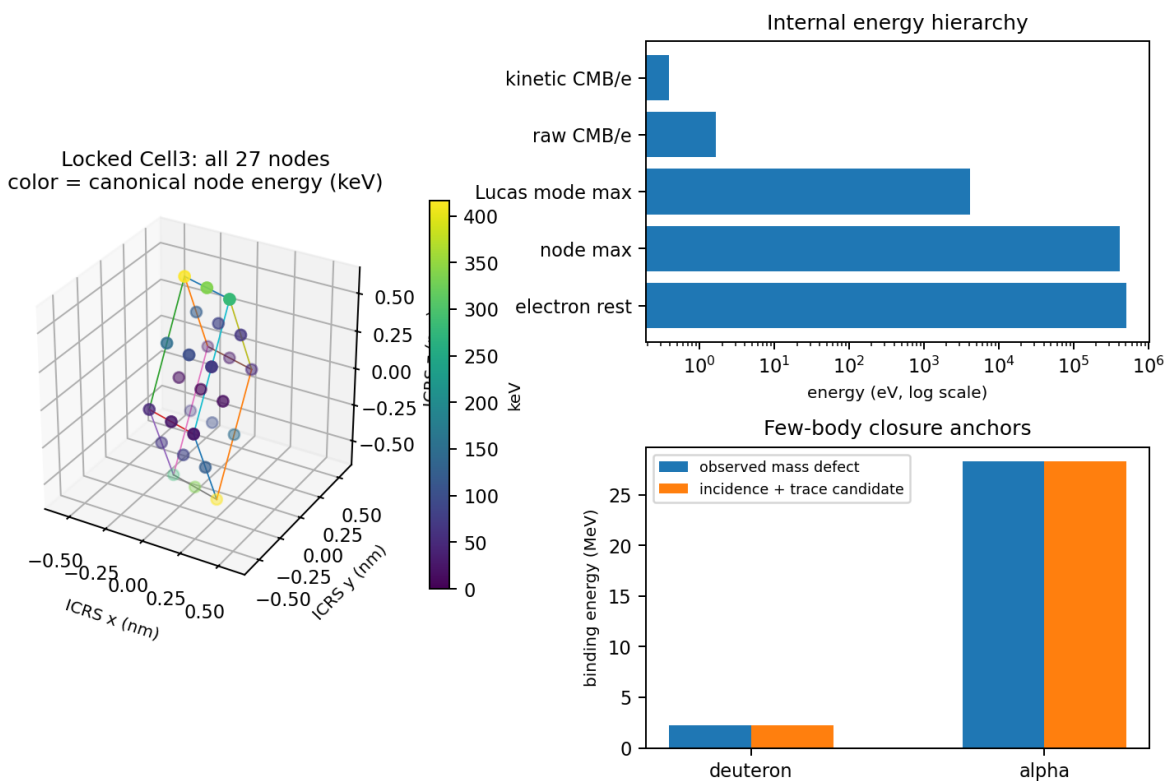


Figure 1. Title composite generated exclusively from the locked Cell3 basis, the complete node-energy ledger, selected Lucas periodic modes, and the deuteron and alpha binding anchors.

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Abstract

This paper develops a complete energy hierarchy for the locked Cell3 and the full Lucas Group. The hierarchy begins with the electron-domain realization scales E_G and L_G , separates the Earth-CMB rate-state energy from rest energy, constructs exact energies on all 27 local Cell3 nodes, derives the 54 nearest-node transition energies, distinguishes the reciprocal rest-mass radials from the direct configuration-energy map, and incorporates the Emily connectivity spectrum, the compact A2 phase energy, quotient-fiber renormalization, 27 Lucas translation-phase sectors, and the 343-node merged Lucas Group. Nuclear mass defect is then placed as the free-to-composite closure deficit of a multi-address state rather than as an isolated face-ratio residual. The deuteron and alpha particle provide the first numerical anchors. All figures and tables are generated from the numerical matrices and constants printed in the archival corpus.

The main structural result is that the K-Field contains several energy readings of the same address system. Rest energy is indexed by reciprocal radial harmonics. Local configuration energy is indexed by the positive quadratic form W . Graph propagation is indexed by the Emily operator. Internal phase activation is indexed by U_K on the A2 torus. Lucas collective modes are indexed by reciprocal supercell integers and Z_3^3 sectors. Earth-CMB motion enters as a mixed rate term. Nuclear binding is the difference between the energy of separately addressable nucleons and the minimum energy of the admissible composite Lucas closure.

Status convention

Class	Meaning
Exact internal	Identity, count, spectrum, or ledger derived directly from the fixed geometry.
Derived dimensional	Exact geometric quantity multiplied by a fixed electron-domain realization scale.
Calibration candidate	Specific proposed mapping to physical energy that is numerically testable.
Theorem target	Operator whose variables and closure role are fixed, but whose universal physical coefficient remains to be established.

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1. Purpose, scope, and dimensional contract

The purpose is to assemble one internally consistent energy map from the complete K-Field corpus while preserving the distinction between geometry, dimensional realization, dynamical coupling, and nuclear closure. A single number cannot serve all four roles. The energy hierarchy is therefore layered rather than collapsed.

The physical Cell3 basis B carries nanometre units. Integer Lucas addresses are dimensionless. The direct metric $M=B^T B$ and the address coupling W carry nm^2 . The reciprocal metric $R=M^{-1}$ carries nm^{-2} . The phase tensor G_{phase} is dimensionless. The Earth traversal rate $qdot$ carries s^{-1} . The electron energy E_G supplies the dimensional energy currency, while L_G supplies the associated length realization.

$$\begin{aligned}x &= (1/2) B q \\M &= B^T B \\R &= M^{-1} \\G_{\text{phase}} &= 4 B^{-T} W B^{-1} \\E_G &= m_e c^2 \\L_G &= |a+b+c|\end{aligned}$$

The report uses four discipline rules. First, the locked Mass-Bridge fixes the carrier shape and is not refitted. Second, the Earth-CMB branch conditions rate-state energy but does not generate the mass anchor. Third, topology fixes mode ratios and degeneracies but requires a constitutive coefficient before it becomes a laboratory energy. Fourth, nuclear binding must be produced as a composition-dependent closure deficit and may not be assigned to a universal one-cell normalization residual.

2. Locked Cell3 geometry and Mass-Bridge carrier

The ordered physical basis is reconstructed in ICRS nanometres from the exact locked native basis and the corpus orientation map. The resulting geometry agrees with the printed ICRS basis to approximately 10^{-11} nm while preserving the exact locked face ratios.

Vector	ICRS x (nm)	ICRS y (nm)	ICRS z (nm)	Length (nm)
a	0.327501242057	-0.054897007885	-0.037487502896	0.334179679060
b	0.035330775842	0.207319850438	-0.547401085391	0.586410890413
c	-0.010129317630	-0.389787702371	-0.689734408368	0.792319765045

Quantity	Value	Unit
angle(a,b)	83.933483012313	degree
angle(a,c)	80.448130018338	degree
angle(b,c)	50.363231813756	degree
A_ab	1.948691654616658e-01	nm^2
A_ac	2.611062357917886e-01	nm^2
A_bc	3.578095391307310e-01	nm^2
volume	1.179148919128169e-01	nm^3
body diagonal L_G	1.343653621679999	nm

Dimensionally correct locked Cell3 with ab, ac, and bc central planes

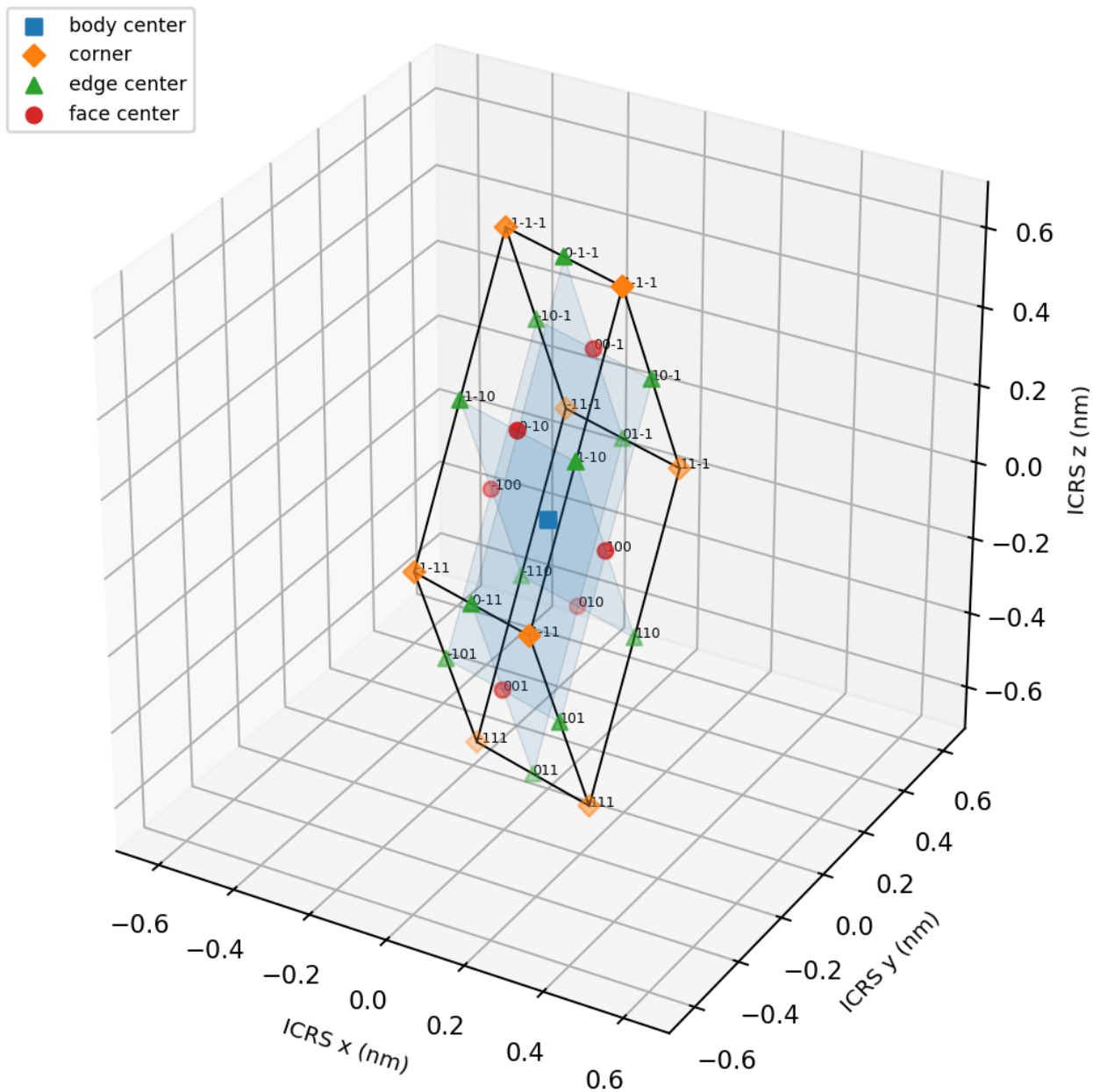


Figure 2. The locked Cell3 in physical ICRS nanometres. All 27 local Lucas nodes, the outer cell edges, and the three central primitive face planes are generated from the ordered basis B.

Central primitive face planes and all in-plane Lucas nodes

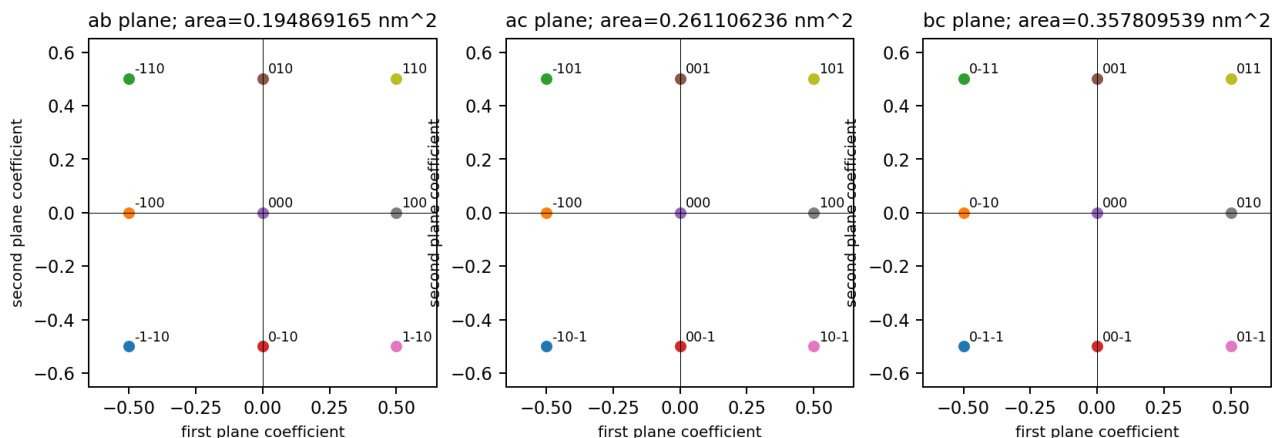


Figure 3. The ab, ac, and bc central planes in primitive coefficient coordinates. Every node lying in each plane is labeled by its centered Lucas address.

The Mass-Bridge is the projective face-area lock

```
1000 A_bc/A_ab = eta = m_p/m_e
100 A_bc/A_ac = alpha^-1
10 A_ac/A_ab = eta alpha
```

Locked relation	Computed value	Authoritative value
1000 A_bc/A_ab	1836.152673425999	1836.152673426000
100 A_bc/A_ac	137.035999177000	137.035999177000
10 A_ac/A_ab	13.399053419929	13.399053419929

The face triangle fixes the proton-electron mass ratio and the electromagnetic coupling coordinate. It provides the free-particle and electromagnetic boundary conditions for the later energy hierarchy. The nuclear binding term enters only after multiple free addresses are composed.

3. Electron realization and Earth-CMB rate-state energy

The electron-domain realization supplies the dimensional energy ruler. E_G is the electron rest energy and L_G is the Cell3 body-diagonal realization length. S_{CMB} links L_G to the electron Compton scale and the local action scale.

```
E_G = 510998.95069 eV
L_G = 1.343653621680 nm
lambda_C,e = L_G/S_CMB
tau_G = L_G/c
h = E_G L_G/(S_CMB c)
```

Layer	Formula	Energy
electron rest realization	E_G	510998.95069 eV
body-diagonal action quantum	E_G/S_{CMB}	922.739286616 eV
electron raw Earth-CMB phase	E_G/S_{CMB}^2	1.666241760 eV
electron normalized Earth-CMB kinetic	$(1/2) E_G \beta_{L^2}$	0.388787039 eV

A physical velocity maps to the half-step rate state $\dot{q}=2B^{-1}v$. The raw phase scalar is anisotropic, but the occupied projective direction supplies the exact self-indexed normalization that returns ordinary kinetic energy.

```
Phi = qdot^T W qdot = v^T G_phase v
C(u) = u^T G_phase u
beta(u) = 1/[2 C(u)]
E_kin/m = beta(u) Phi = |v|^2/2
```

Quantity	Value
Earth-CMB vector	(-359003.199066, 76679.886034, -44701.060101) m/s

Quantity	Value
speed	369812.502000406 m/s
directional phase cost C_E	2.142872054067834
raw phase scalar	293.061936693966 GJ/kg
normalization beta_E	0.233331711682871
normalized kinetic energy	68.380643317900 GJ/kg

The Earth-CMB state therefore occupies a low-energy rate layer inside the electron realization. It does not replace or generate the rest-energy ladder. Its physical role is to condition traversal, transition timing, and mixed local-motion terms.

4. Complete local Cell3 node-energy map

Each local address $L=(i,j,k)$ belongs to $\{-1,0,1\}^3$. The exact positive configuration index is $U_L=(1/2)L^T W L$. It has nm^2 units because W carries nm^2 . The simplest dimensionalization using only the electron realization scales is $\kappa_0=E_G/L_G^2$. This coefficient is a canonical ruler rather than a completed nuclear coupling.

$$U_L = (1/2) L^T W L$$

$$\kappa_0 = E_G/L_G^2 = 0.283038395877 \text{ MeV/nm}^2$$

$$E_L(\theta) = \kappa_0 U_L$$

Canonical dimensionalized node-energy map: $E_L=(E_G/L_G^2)(1/2 L^T W L)$

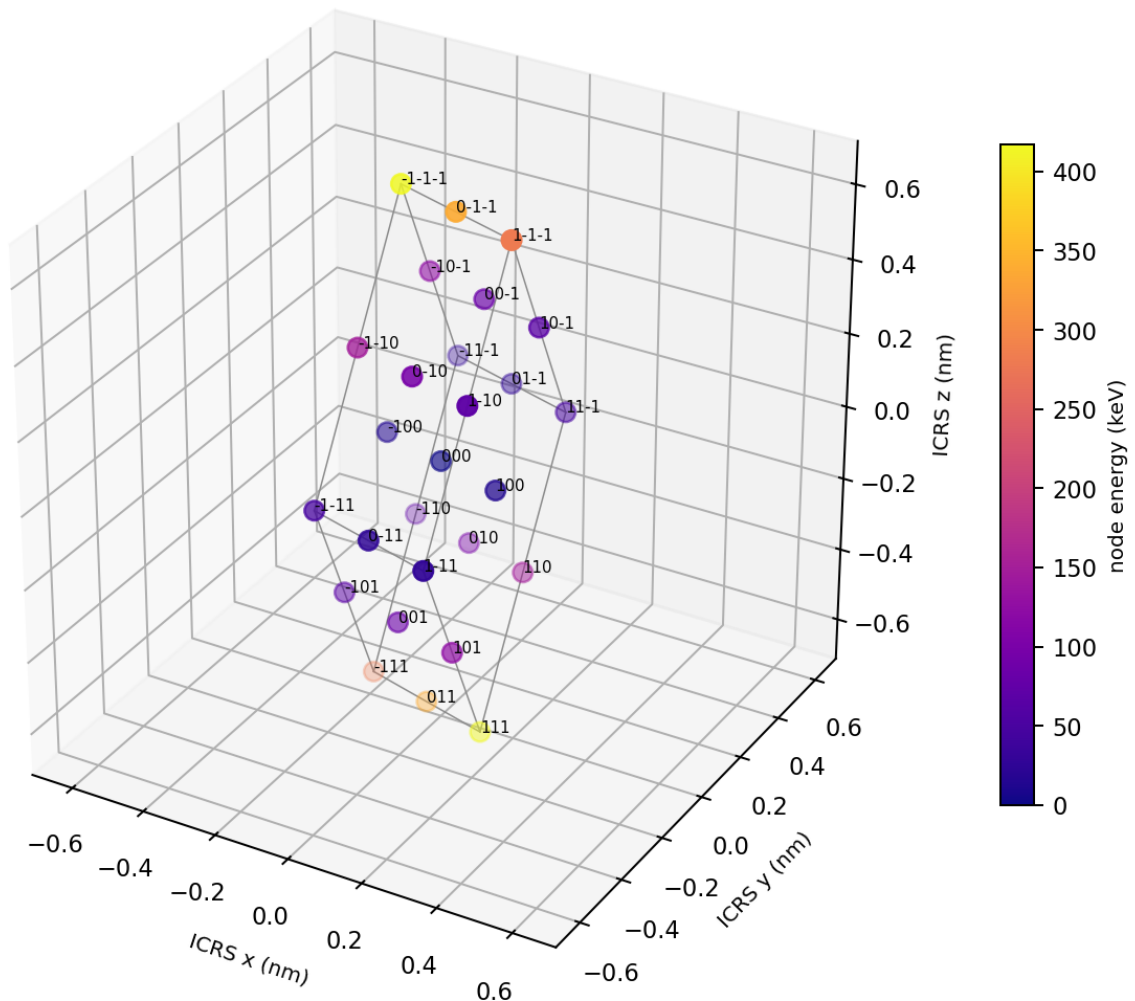


Figure 4. Exact local node-energy index dimensionalized by kappa_0. Antipodal nodes have identical scalar energy; the directed address remains distinct.

Address pair	Class	lambda	U_L (nm^2)	E_L^(0) (keV)
+/- 100	primitive a-axis	1	0.041676651	11.796093
+/- 01-1	bc face diagonal (-)	0	0.089272943	25.267671
+/- 1-11	body diagonal (+++)	1	0.096008229	27.174015
+/- 11-1	body diagonal (++-)	1	0.165890960	46.953511
+/- 10-1	ac face diagonal (-)	0	0.232991266	65.945474
+/- 1-10	ab face diagonal (-)	0	0.250376096	70.866049
+/- 001	primitive c-axis	1	0.293873826	83.177576
+/- 010	primitive b-axis	1	0.346200021	97.987899
+/- 101	ac face diagonal (+)	2	0.438109689	124.001864
+/- 110	ab face diagonal (+)	2	0.525377249	148.701934
+/- 1-1-1	body diagonal (++-)	-1	0.992491616	280.913235
+/- 011	bc face diagonal (+)	2	1.190874752	337.063280
+/- 111	body diagonal (+++)	3	1.472611191	416.805509

The complete 27-node ledger appears in Appendix B. The center has zero configuration energy. The thirteen nonzero antipodal pairs span approximately 11.8 to 416.8 keV under kappa_0. These values are local configuration energies, not particle rest masses.

5. Seven-sheet energy hierarchy and six-transition closure

The sheet index $\lambda = i+j+k$ takes seven values from -3 to +3. The sheet populations are the coefficients of $(z^{-1}+1+z)^3$ and are exactly 1,3,6,7,6,3,1. Alternating polarity $\chi = (-1)^\lambda$ supplies a Z_2 grading. The A2 registry supplies a Z_3 grading. Together they form Z_6, so six primitive transitions return the combined state to its origin and seven endpoint sheets display one complete interval.

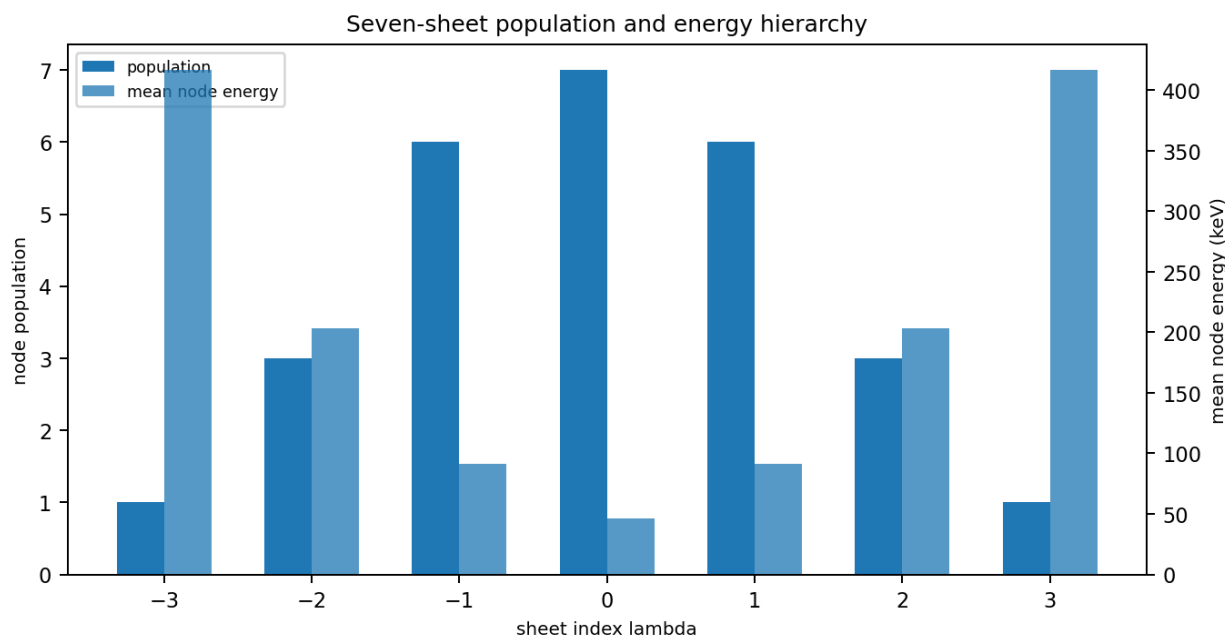


Figure 5. Seven-sheet population and mean node-energy hierarchy. The central sheet contains the most nodes but the lowest mean quadratic cost.

lambda	Population	Polarity	Total U (nm^2)	Mean U (nm^2)	Mean energy (keV)
-3	1	-1	1.472611191	1.472611191	416.805509
-2	3	1	2.154361690	0.718120563	203.255692
-1	6	-1	1.936141303	0.322690217	91.333721
0	7	1	1.145280611	0.163611516	46.308341

lambda	Population	Polarity	Total U (nm^2)	Mean U (nm^2)	Mean energy (keV)
1	6	-1	1.936141303	0.322690217	91.333721
2	3	1	2.154361690	0.718120563	203.255692
3	1	-1	1.472611191	1.472611191	416.805509

Sheet grading separates external progression from same-sheet transport. A transition e_i changes lambda by one and flips polarity. A root transition e_i - e_j preserves lambda and polarity while moving in the A2 fiber. This is the first separation between externally visible sheet energy and internally retained phase energy.

6. Nearest-node transition energy and discrete gradients

The local $P3 \times P3 \times P3$ graph contains 54 undirected nearest-node links. The exact forward and backward energy increments are obtained algebraically from W . The common self-term cancels from the centered difference, leaving an address covector W_L .

$$\begin{aligned}\Delta_{i^+} U(L) &= (W_L)_i + w_{ii}/2 \\ \Delta_{i^-} U(L) &= -(W_L)_i + w_{ii}/2 \\ \delta_i U(L) &= [U(L+e_i) - U(L-e_i)]/2 = (W_L)_i\end{aligned}$$

The physical finite-difference vector is obtained through B^T . For the quadratic law it agrees exactly with the continuous ellipsoidal normal evaluated at the resolved node.

$$\begin{aligned}g_D(L) &= -2 \kappa_W B^T W_L \\ r_L &= B L/2 \\ g_D(L) &= -\kappa_W G_{\text{phase}} r_L\end{aligned}$$

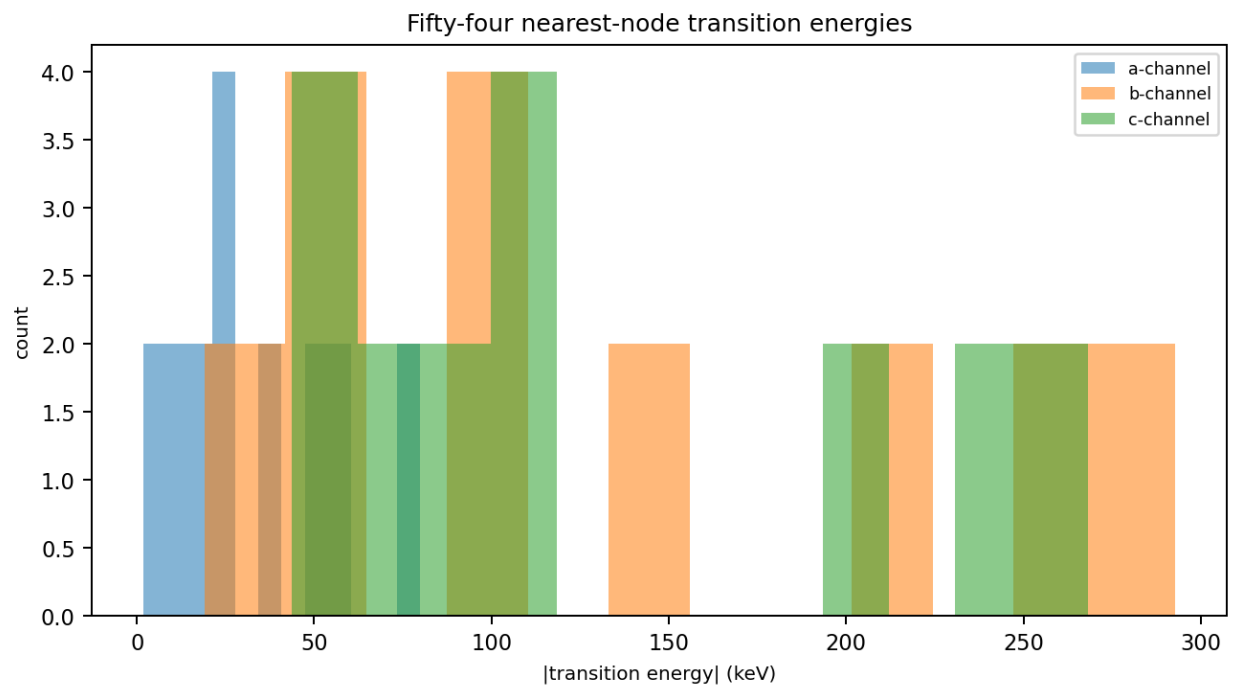


Figure 6. Distribution of all 54 nearest-node transition energies under κ_0 . Channel asymmetry is produced by the unequal diagonal and cross terms of W .

Channel	Minimum $ \Delta E $	Maximum $ \Delta E $	Link count
a	1.906345 keV	79.742230 keV	18
b	18.991963 keV	292.803646 keV	18
c	43.692033 keV	268.103575 keV	18

The complete transition ledger appears in Appendix C. It is essential for experiments because spectroscopic observations couple to differences between states rather than to absolute node energies alone.

7. Reciprocal rest-energy radial hierarchy

The reciprocal metric R produces thirteen projective radial units on the nonzero $\{-1,0,1\}^3$ shell after antipodal pairing. For address k , $q(k)=k^T R k$ and $s_k=\sqrt{q(k)/q(001)}$. Particle rest energies occupy integer or subharmonic positions on these radial ladders.

$$\begin{aligned} q(k) &= k^T R k \\ s_k &= \sqrt{q(k)/q(001)} \\ E_{\text{rest}}(k, p, d) &= E_G(p/d) s_k \end{aligned}$$

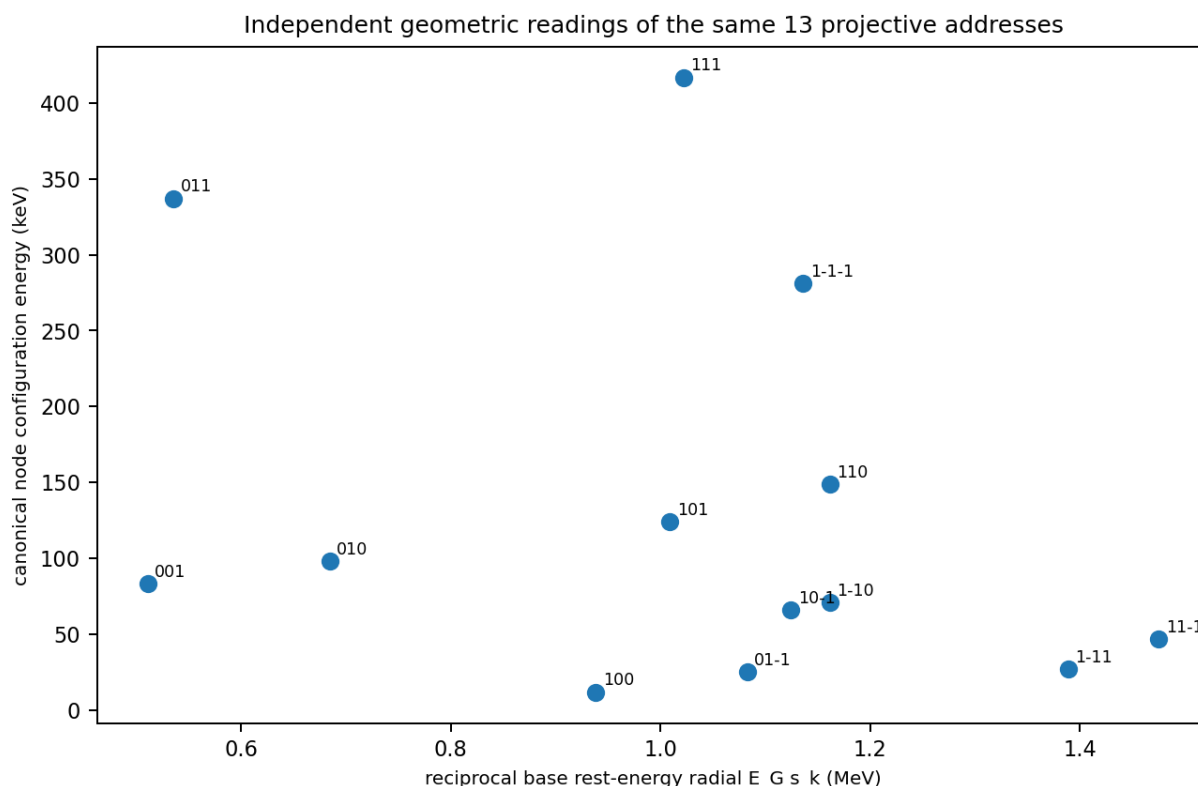


Figure 7. Reciprocal base rest-energy radials versus direct local configuration energies. Their nonmonotonic relationship proves that the two maps are independent readings of one address grammar.

Radial	Description	q (nm ⁻²)	s_k	$E_G s_k$ (MeV)	$E_{\text{node}}^{(0)}$ (keV)
001	primitive c-axis	2.731171275	1.000000000000	0.510998951	83.177576
011	bc face diagonal (+)	3.004218418	1.048796632266	0.535933979	337.063280
010	primitive b-axis	4.903398312	1.339905341993	0.684690224	97.987899
100	primitive a-axis / Mass-Bridge	9.208025529	1.836152673426	0.938272089	11.796093
101	ac face diagonal (+)	10.648820733	1.974587075075	1.009011923	124.001864
111	body diagonal (+++)	10.924937481	2.000023101888	1.022009706	416.805509
01-1	bc face diagonal (-)	12.264920756	2.119131490765	1.082873968	25.267671
10-1	ac face diagonal (-)	13.229572875	2.200890538669	1.124652756	65.945474
1-1-1	body diagonal (+--)	13.499550414	2.223234046131	1.136070265	280.913235
1-10	ab face diagonal (-)	14.108354237	2.272813025845	1.161405071	70.866049
110	ab face diagonal (+)	14.114493447	2.273307476082	1.161657735	148.701934
1-11	body diagonal (+-+)	20.179500609	2.718195812855	1.388995208	27.174015
11-1	body diagonal (++-)	22.766391961	2.887171860755	1.475341791	46.953511

The electron is the 001 unit state. The proton is the thousandth harmonic of the 100 Mass-Bridge radial. The neutron occupies the near-double 111 radial. The rest-energy ladder therefore sits above the local node-energy map and uses a different metric object.

8. Emily connectivity modes

The Emily operator is the adjacency matrix of the finite local graph $P3 \times P3 \times P3$. It is metric-independent. Each one-dimensional $P3$ factor has eigenvalues $-\sqrt{2}$, 0, and $+\sqrt{2}$, so every three-axis eigenvalue is a sum of three such symbols.

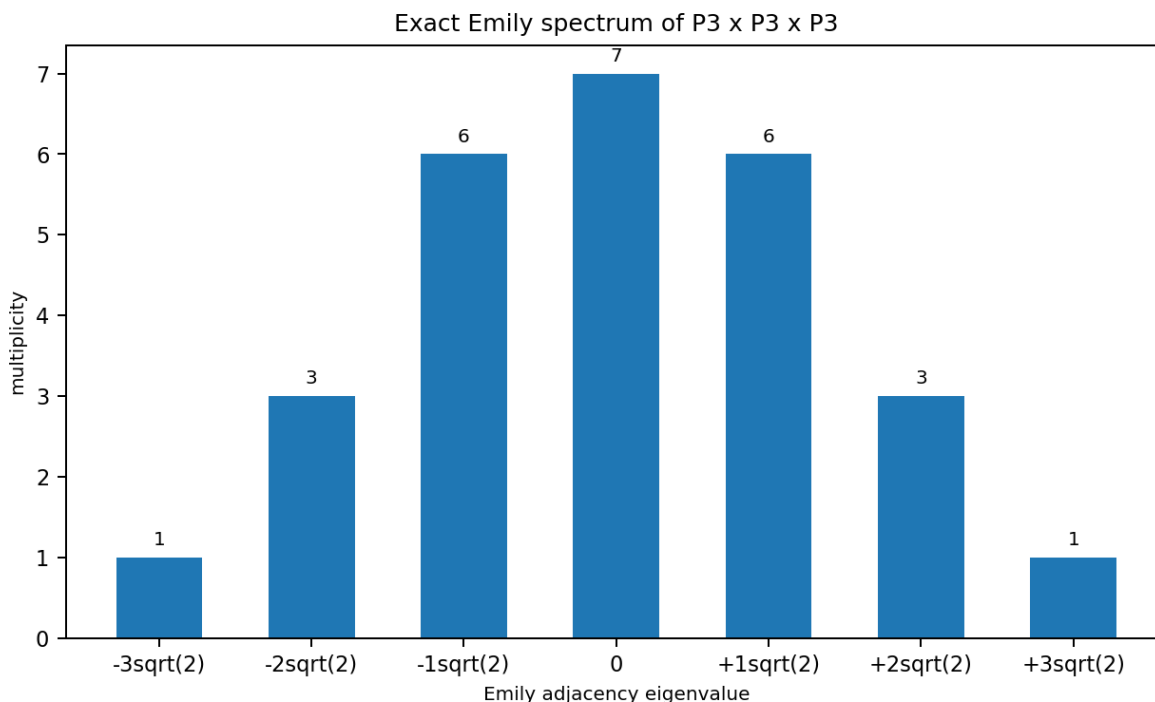


Figure 8. Exact Emily adjacency spectrum. The multiplicity sequence is the same 1,3,6,7,6,3,1 sequence as the seven-sheet populations because both arise from a threefold convolution.

$$A_L = A_{P3} \text{ tensor } I \text{ tensor } I + I \text{ tensor } A_{P3} \text{ tensor } I + I \text{ tensor } I \text{ tensor } A_{P3}$$

$$\lambda_{\text{Emily}} = (m_a + m_b + m_c) \sqrt{2}$$

A physical coupling J_E converts this exact topology into an energy spectrum: $H_{\text{Emily}} = E_c I - J_E A_{\text{Emily}}$. The level ratios and degeneracies are fixed; the absolute splitting J_E must be derived from a constitutive law or measured.

The seven-dimensional zero eigenspace has the same dimension as the seven compatibility syndromes in the thirteen-channel quadratic metric. This shared dimension identifies the central Emily sector as a natural carrier for nonmetric residual content, closure constraints, or internally balanced modes.

9. Compact A2 phase energy and internal mode stiffness

The same-sheet roots close as $s_{ac} = s_{ab} + s_{bc}$, leaving two independent loop phases. The exact compact energy therefore lives on a two-dimensional A2 character torus.

$$U_K(\psi_1, \psi_2) = w_{ab}[1 - \cos \psi_1] + w_{bc}[1 - \cos \psi_2] + w_{ac}[1 - \cos(\psi_1 + \psi_2)]$$

Channel	Weight	Share
ab	0.396285985688453	0.051818%
ac	166.587550282441185	21.782922%
bc	597.778333979951640	78.165259%

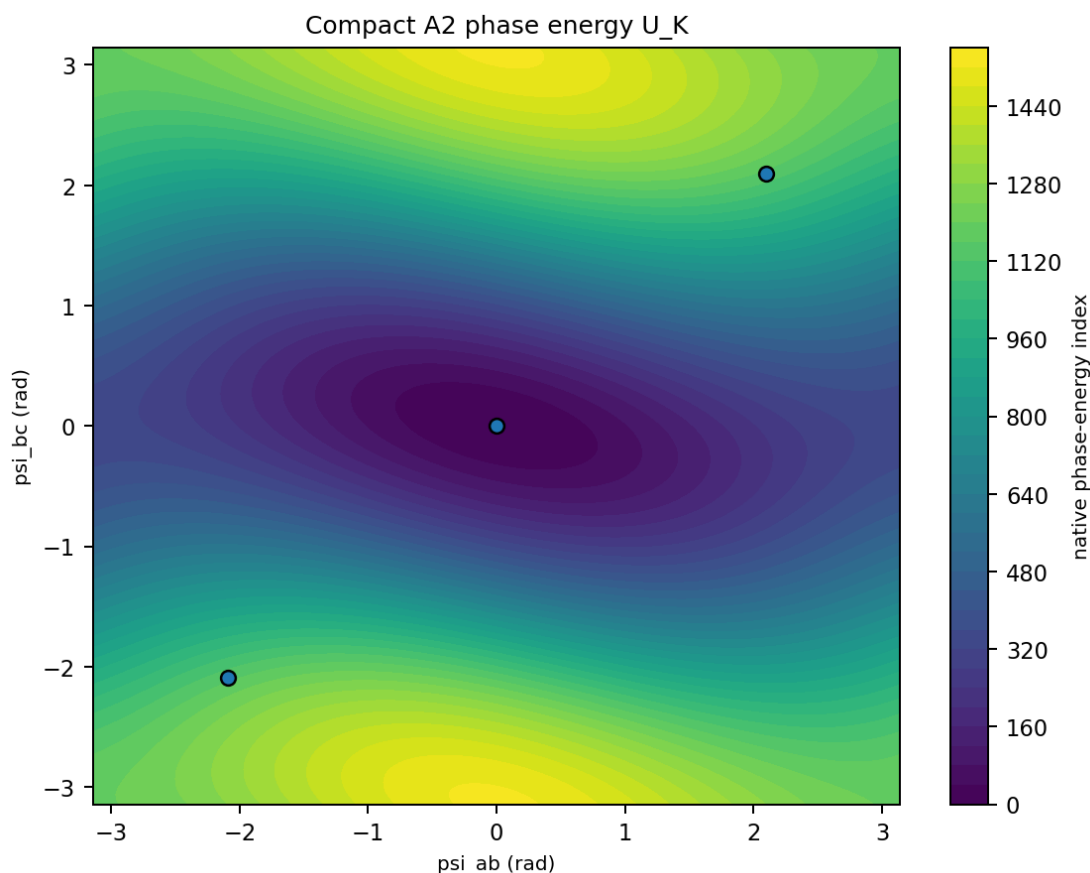


Figure 9. Compact A2 phase energy. The cold equal-phase registry is at the origin. The two conjugate nonzero Z_3 registries are finite hot states.

$$U_{\text{hot}} = (3/2)(w_{\text{ab}} + w_{\text{ac}} + w_{\text{bc}}) = 1147.143255372122$$

Object	Eigenvalues
weighted branch Laplacian	-0.000000000, 230.716133122, 1298.808207374
cold-state two-phase Hessian	123.669392874, 807.680327657

The scalar U_K is even under phase conjugation, its gradient is odd, and its Hessian is even. One generating energy therefore supplies three readings: scalar activation, oriented response, and local stiffness. This hierarchy is the natural internal bridge among binding, electromagnetic orientation, and material or collective response.

10. Quotient-fiber energy renormalization

The Cell3 metric admits an exact one-plus-two coordinate split into a coherent sheet-normal coordinate σ and two A2 registry coordinates ξ . The algebraic subspaces are distinct, but the physical metric contains nonzero cross blocks. Eliminating the internal coordinates therefore changes the apparent coherent cost.

$$\begin{aligned} \rho_{\text{QF}}^2 &= c^T F_M^{-1} c / m_0 \\ m_{\text{eff}} &= m_0(1 - \rho_{\text{QF}}^2) \\ \chi_{\text{QF}} &= 1/(1 - \rho_{\text{QF}}^2) \end{aligned}$$

Quantity	Value
ρ_{QF}^2	0.543701653538894
$1 - \rho_{\text{QF}}^2$	0.456298346461106
m_0	0.601801685017926 nm ²
m_{eff}	0.274601113771187 nm ²
χ_{QF}	2.191548594807890

Quotient-fiber energy pathway

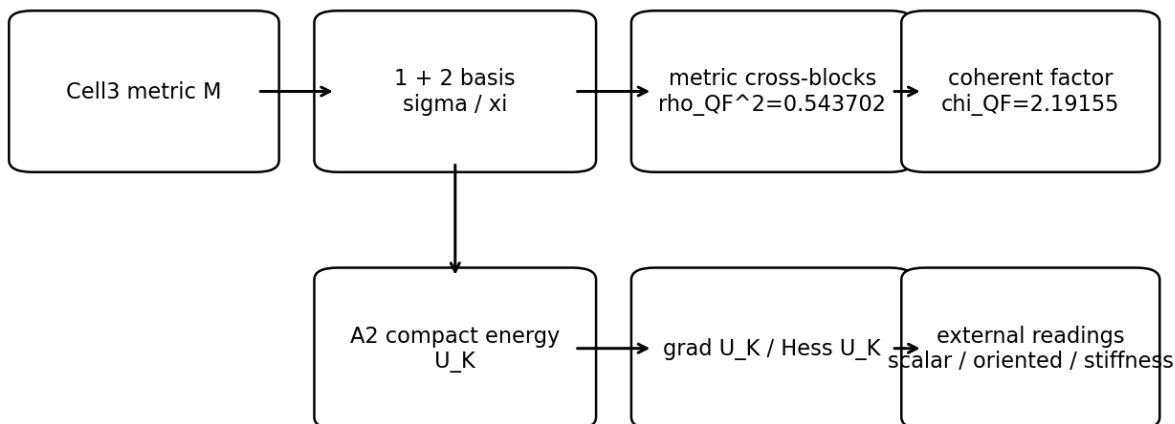


Figure 10. Quotient-fiber pathway. Common transport, internal A2 phase, and external readings are coupled by the Cell3 metric rather than being independent layers.

The factor $\chi_{\text{QF}}=2.19155$ is the exact metric cost of treating coherent transport as if internal registry were absent. For a bound composite, this provides a geometric mechanism by which an externally measured mass projection can be lower while finite energy remains stored in internal phase and fiber coordinates.

11. Full Lucas Group geometry and collective closure

The Lucas Group is the $3 \times 3 \times 3$ block of Cell3 cells centered on one selected cell. Its 27 cell centers form the finite translation group Z_3^3 . Merging the local 27-node grids gives the global half-step addresses L in $\{-3,-2,-1,0,1,2,3\}^3$ and therefore 343 unique nodes.

Full 3 x 3 x 3 Lucas Group: 343 merged half-step nodes

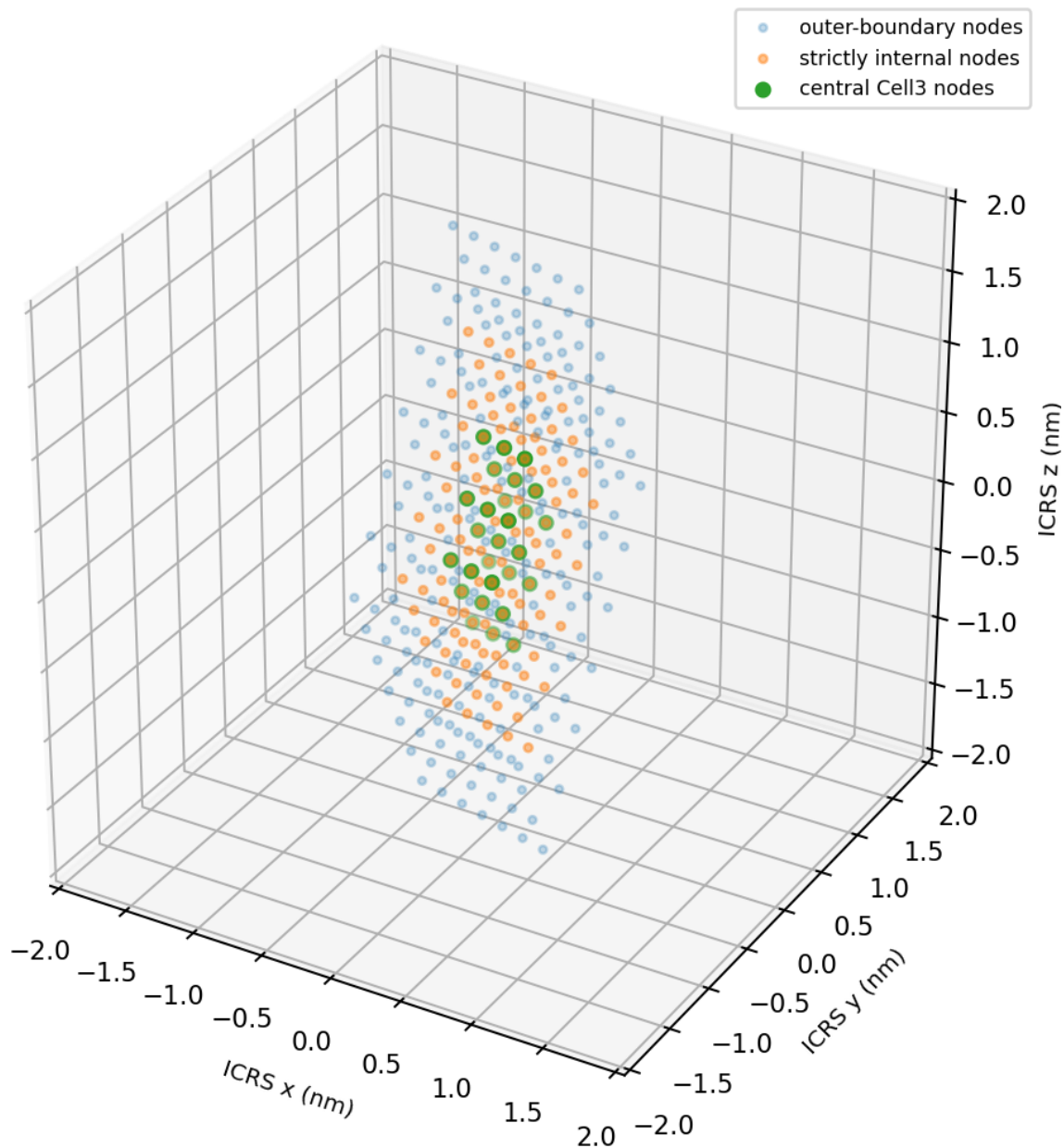


Figure 11. Complete Lucas Group. The 343 merged nodes include 125 strictly internal points and 218 outer-boundary points. Every node of the central Cell3 is internal to the group.

Object	Count
Cell3 cells	27
cell-center phase sectors	27
merged half-step nodes	343
strictly internal nodes	125
outer-boundary nodes	218
internal shared primitive faces	54
exposed primitive faces	54
global laminar sheets	19

At $N=3$ the internal and exposed primitive face counts are equal in every face family: 18 internal and 18 exposed ab faces, 18 and 18 ac faces, and 18 and 18 bc faces. This is the first finite assembly in which internal interface expression and external boundary

expression balance exactly.

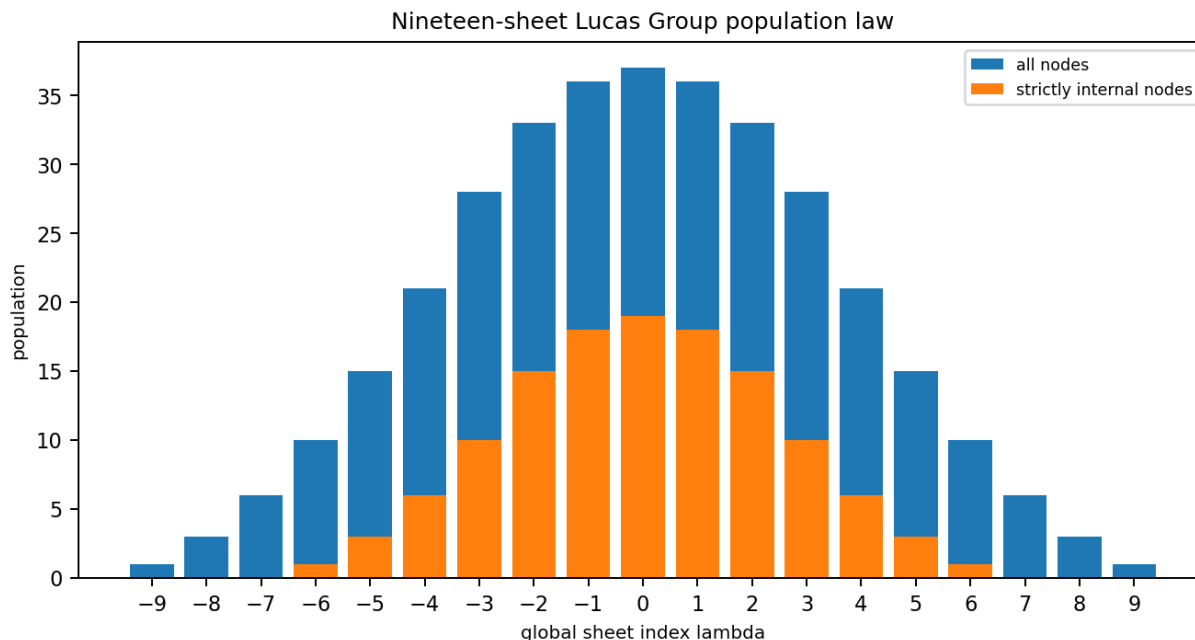


Figure 12. Nineteen-sheet population law of the 343-node group. The central sheet contains 37 nodes, including 19 strictly internal nodes.

The global node-energy trace supplies an occupation-independent normalization budget. For addresses $L_i=-r, \dots, +r$, the exact sum is

$$\text{sum } U_L = [r(r+1)(2r+1)^{3/6}] \text{ trace}(W)$$

Window	r	Node count	sum U_L	kappa_0 sum U_L
local Cell3	1	27	12.271508981	3.473308217 MeV
full Lucas Group	3	343	935.361684515	264.743270750 MeV

These totals are exact geometry-normalization budgets, not automatically occupied physical energies. A physical model must specify which nodes, links, or collective modes are populated.

12. Lucas periodic modes and twenty-seven Bloch sectors

For the outer basis $C=3B$, an integer supercell mode N has reciprocal magnitude squared $(1/9)N^T R N$. The exact wavelength, frequency, and action quantum are

$$\begin{aligned} \lambda_{N} &= 3/\sqrt{N^T R N} \\ f_N &= (c/3)\sqrt{N^T R N} \\ E_N &= h f_N = [E_G L_G / (3 S_{CMB})]\sqrt{N^T R N} \end{aligned}$$

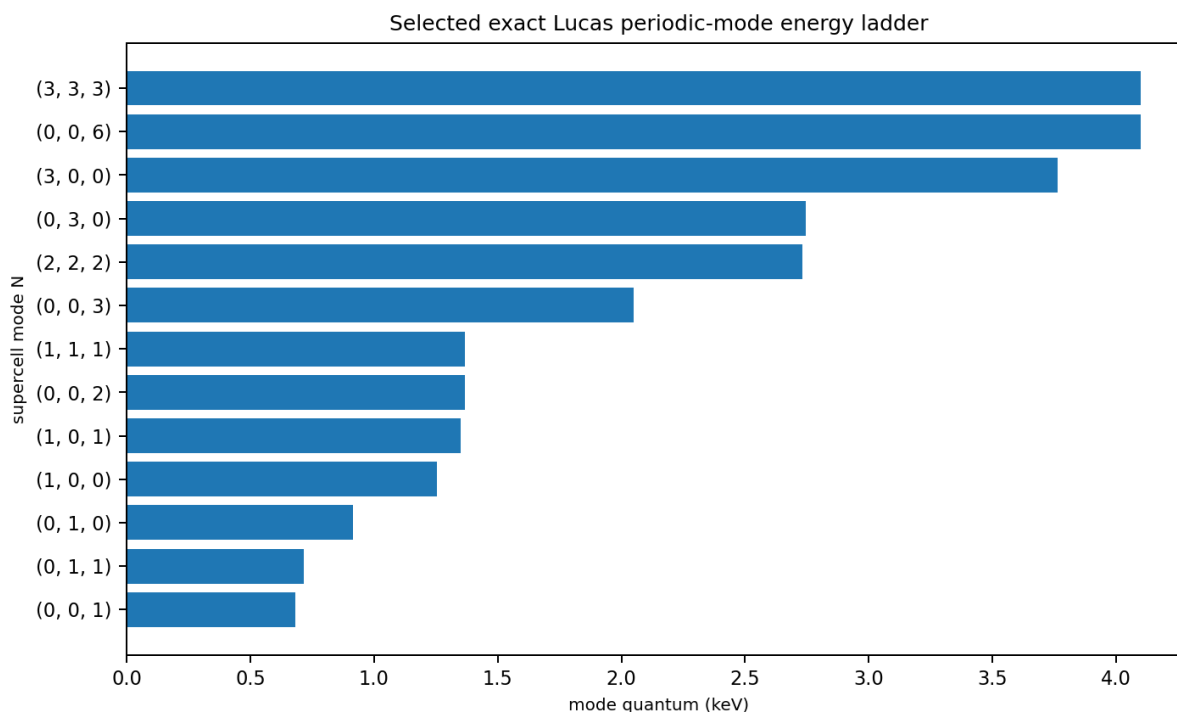


Figure 13. Selected Lucas periodic-mode energies. The ordering is fixed by the oblique reciprocal metric, not by independent edge lengths.

N	Sector mod 3	Role	Wavelength (nm)	Frequency (PHz)	Energy
(0, 0, 1)	[0, 0, 1]	lowest listed c-like supercell mode	1.815293224561	165.148227147	0.682998 keV
(0, 1, 1)	[0, 1, 1]	low mixed mode	1.730834337863	173.206904456	0.716326 keV
(0, 1, 0)	[0, 1, 0]	b-like supercell mode	1.354792139168	221.282991774	0.915153 keV
(1, 0, 0)	[1, 0, 0]	a-like supercell mode	0.988639589089	303.237358787	1.254089 keV
(1, 0, 1)	[1, 0, 1]	mixed mode	0.919328019248	326.099554795	1.348639 keV
(0, 0, 2)	[0, 0, 2]	near-degenerate diagonal partner	0.907646612281	330.296454293	1.365996 keV
(1, 1, 1)	[1, 1, 1]	diagonal fundamental	0.907636128226	330.300269529	1.366012 keV
(0, 0, 3)	[0, 0, 0]	c-axis Lucas-grid nodal family	0.605097741520	495.444681440	2.048995 keV
(2, 2, 2)	[2, 2, 2]	second diagonal harmonic	0.453818064113	660.600539058	2.732024 keV
(0, 3, 0)	[0, 0, 0]	b-axis Lucas-grid nodal family	0.451597379723	663.848975323	2.745459 keV
(3, 0, 0)	[0, 0, 0]	a-axis Lucas-grid nodal family	0.329546529696	909.712076360	3.762267 keV
(0, 0, 6)	[0, 0, 0]	parity partner	0.302548870760	990.889362879	4.097989 keV
(3, 3, 3)	[0, 0, 0]	Lucas parity mode	0.302545376075	990.900808587	4.098036 keV

Translation by one Cell3 multiplies a mode by a cube root of unity. The residue $N \bmod 3$ labels one of the 27 exact Z_3^3 phase sectors. The Gamma sector repeats identically from cell to cell. Nonzero residues introduce plus or minus 120-degree phase progression. The diagonal sequence (1,1,1), (2,2,2), and (3,3,3) produces six-state, three-state, and two-state sheet cycles, respectively.

A boundary model must be specified before assigning a measured resonance. A periodic supercell, a monolithic hard-wall cavity, a partitioned 27-cell device, and a finite-coupling network have different physical spectra even though they share the same underlying geometric index system.

13. Earth-CMB coupling as a transition-rate term

For local motion $\dot{q}_R$ superposed on Earth translation $\dot{q}_E$, the raw phase scalar separates into an Earth baseline, a local self-term, and a mixed interaction.

$$\begin{aligned}\Phi_{\text{total}} &= \dot{q}_E^T W \dot{q}_E + \dot{q}_R^T W \dot{q}_R + 2 \dot{q}_R^T W \dot{q}_E \\ \epsilon_{\text{cross}} &= \kappa_F \dot{q}_R^T W \dot{q}_E\end{aligned}$$

The CMB contribution therefore belongs on transitions, currents, or link phases. It should not be inserted as a universal static offset in the node-energy ledger. A discrete link formulation can attach a phase proportional to $(L_{\text{prime}} - L)^T W \dot{E}$ to every transition.

Branch	$\dot{E} \text{ (s}^{-1}\text{)}$	$(W \dot{E})_i \text{ (m}^2\text{/s)}$
a	-2.217734842949e+15	-1.451788600612e-04
b	2.499065863121e+14	-1.033633529403e-04
c	5.181751775932e+13	-5.934473800887e-05

The branch biases are unequal and signed. This is the event-level origin of direction-dependent local coupling. In the nuclear hierarchy, the same structure can condition transitions between admissible composite states, but it remains distinct from the static binding baseline.

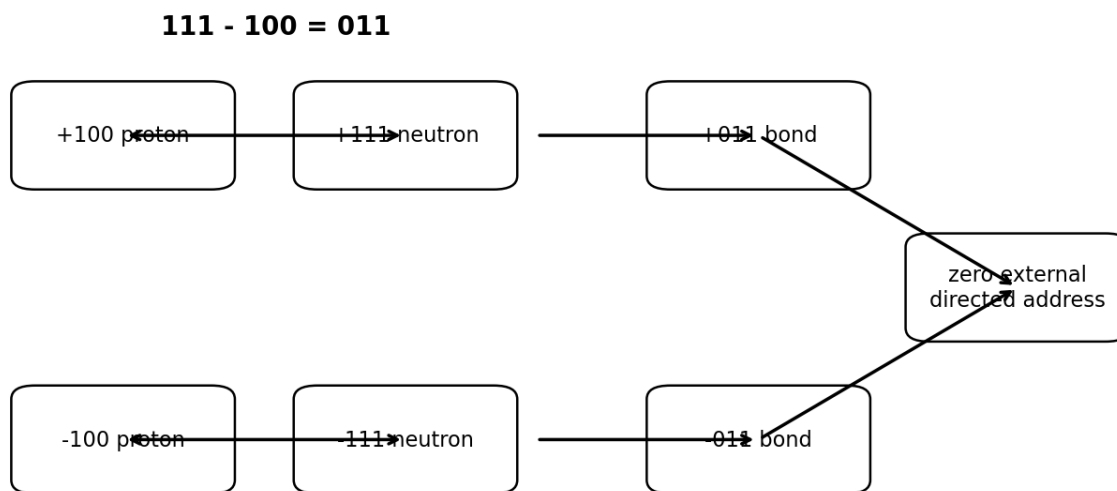
14. Nuclear binding as a Lucas closure deficit

The free nucleon state is represented by separately addressable proton and neutron rest-energy channels. A bound nucleus is represented by a composite address $C^*(Z, N)$ that closes simultaneously in direct space, reciprocal phase, parity, occupancy, and the Lucas translation sectors. The observed binding energy is the difference between these two projections.

$$\begin{aligned} \Delta m(Z, N) &= Z m_p + N m_n - M(Z, N) \\ B(Z, N) &= \Delta m c^2 \\ B_K(Z, N) &= E_{\text{free}}(\{C_i\}) - \min E_{\text{composite}}(C^*(Z, N)) \end{aligned}$$

The binding operator must be finite range, attractive for compatible addresses, repulsive or exclusionary at the inner core, saturating with increasing nucleon count, sensitive to shell closure, and exactly conserving energy, momentum, angular momentum, charge/parity, and occupancy. These requirements locate the missing mass defect in the multi-address closure operator rather than in the locked single-cell face ratios.

Proton-neutron and alpha-type address closure



Two antipodal pn bond channels provide the first complete $2p+2n$ closure.

Figure 15. Proton-neutron and alpha-type address closure. The proton 100 and neutron 111 channels differ by the exact 011 face-diagonal bond channel. Two antipodal pn bonds close the directed $2p+2n$ address.

The direct p-n configuration relation is

$$\begin{aligned} \Delta U_{pn} &= U_{100} + U_{111} - U_{011} = 0.323413090459 \text{ nm}^2 \\ \Delta E_{pn}(\theta) &= \kappa_0 \Delta U_{pn} = 0.091538322329 \text{ MeV} \end{aligned}$$

This direct value is not the complete deuteron binding. Multiplying by the W stiffness contrast χ_W produces a specific zero-parameter candidate of 2.219783 MeV, within approximately 4.783 keV of the CODATA-derived deuteron binding energy. The relation is an experimental calibration candidate and must be tested against additional few-body states.

15. Deuteron and alpha numerical anchors

The corpus carries the following electron-mass-unit residuals: $B_d/E_G=4.353367771$ and $B_{\alpha}/E_G=55.373129142$. Multiplication by E_G gives 2.224566363 MeV and 28.295610888 MeV. These values anchor the dimensional nuclear hierarchy.

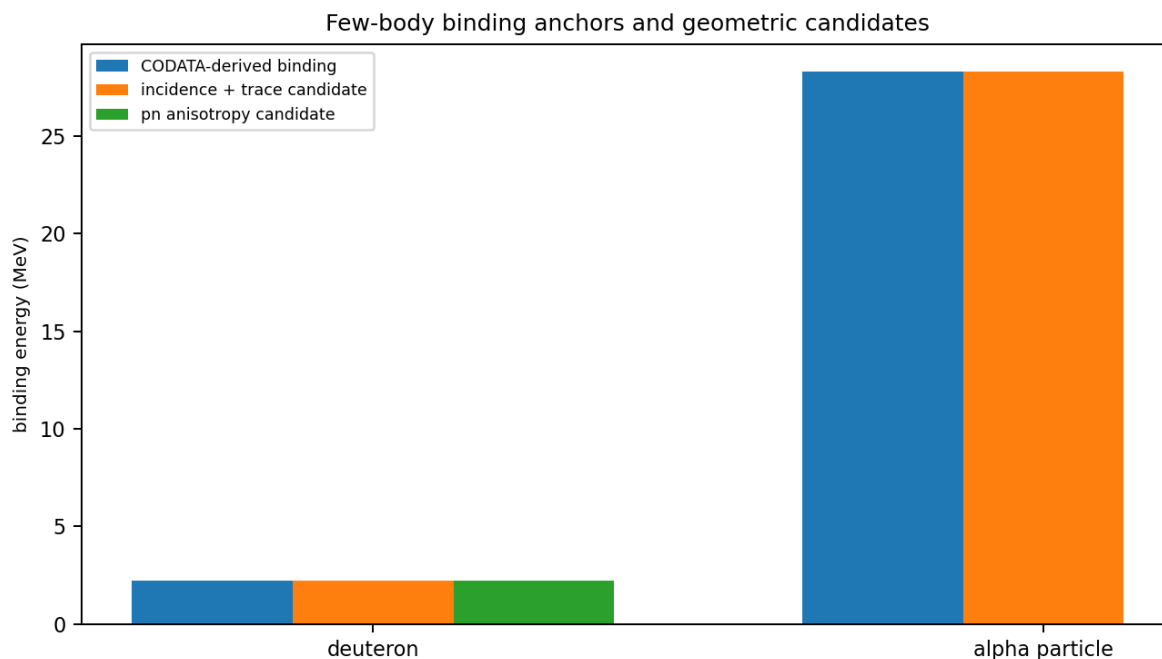


Figure 16. Observed few-body binding energies and the two geometric candidate relations developed from the fixed Cell3 and Lucas ledgers.

System	Observed B	B/E_G	Candidate	Candidate B	Residual
deuteron	2.224566362949 MeV	4.353367771000	3 + trace(W)	2.229744431229 MeV	+5.178068 keV
deuteron	2.224566362949 MeV	4.353367771000	$\chi_W \kappa_0 \Delta U_{pn}$	2.219783160930 MeV	-4.783202 keV
alpha particle	28.295610887984 MeV	55.373129142000	54 + trace(W)	28.290690916419 MeV	-4.919972 keV

The incidence-plus-trace pattern suggests the provisional law $B/E_G = I_{\text{lock}} + \text{trace}(W) + \Delta \text{phase}$. The deuteron uses the minimal three-family closure integer $I_{\text{lock}}=3$. The alpha particle uses the complete Lucas interface count $I_{\text{lock}}=54$. The remaining residual is at the few-keV scale and is a candidate carrier for detailed parity, reciprocal-phase, finite-size, Coulomb, or boundary corrections.

The law is not accepted by numerical proximity alone. It becomes physically meaningful only if I_{lock} is generated independently from address incidence for triton, helion, and heavier nuclei, and if the phase residual is predicted from the same universal operator without nucleus-specific fitting.

16. Master Cell3-Lucas Hamiltonian

The complete working operator is the sum of distinct but coupled layers. Each term has a fixed geometric role and a separate coefficient status.

Master Cell3-Lucas energy operator and projection chain

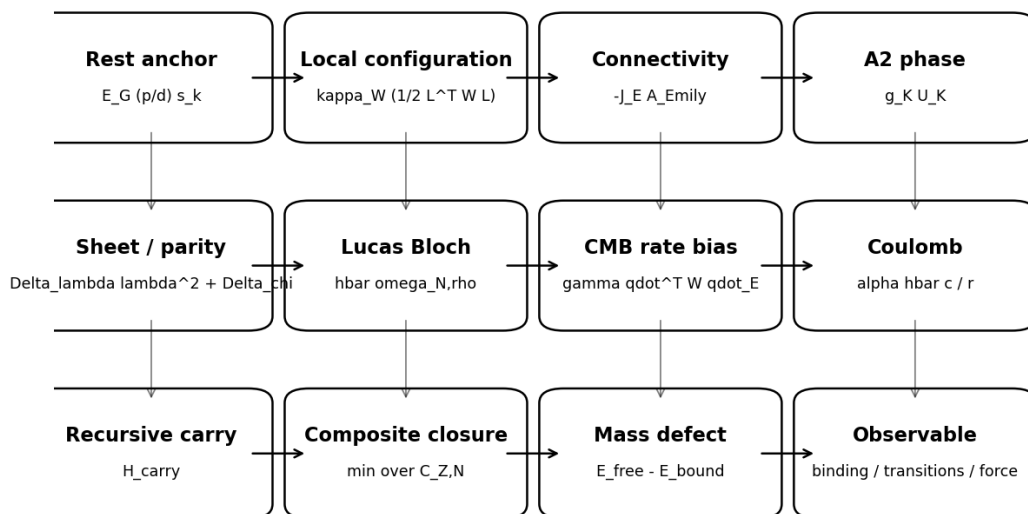


Figure 17. Master energy operator and projection chain from the electron rest anchor to nuclear mass defect and observable transitions.

$$H_{\text{Lucas}} = H_{\text{rest}} + H_W + H_{\text{Emily}} + H_{\text{A2}} + H_{\text{sheet}} + H_{\text{Bloch}} + H_{\text{CMB}} + H_{\text{Coulomb}} + H_{\text{carry}}$$

Term	Form	Role	Status
H _{rest}	$\sum E_G(p/d) s_k n_k$	particle rest-energy radials	exact radial grammar plus observed assignments
H _W	$\kappa_W \sum (1/2 L^T W L) n_L$	local configuration energy	exact index; coefficient to calibrate
H _{Emily}	$-J_E \sum \text{nearest-link hopping}$	connectivity propagation	exact ratios; J _E open
H _{A2}	$g_K U_K(\psi_1, \psi_2)$	internal registry phase energy	exact form; g _K open
H _{sheet}	$\Delta_\lambda \lambda^2 + \Delta_\chi(1-\chi)/2$	sheet and polarity splitting	operator target
H _{Bloch}	$\sum \hbar \omega_{N,\rho} (b^\dagger b + 1/2)$	Lucas collective modes	exact periodic spectrum after boundary declaration
H _{CMB}	$\gamma \dot{q}^T W \dot{q}_E$	rate-state bias and mixed motion	motion branch coefficient
H _{Coulomb}	$\alpha \hbar c \sum_{pp} P_{ij} / r_{ij}$	proton repulsion	known far-field target
H _{carry}	recursive carry and motif energy	cross-level closure transitions	theorem target

The nuclear prediction is obtained by minimizing the non-rest part of H_{Lucas} over all states satisfying the fixed proton and neutron counts, direct and reciprocal syndromes, parity, Bloch-sector consistency, short-range compatibility, and conservation laws.

$$B_K(Z, N) = E_{\text{free}}(Z, N) - \min_{\{\Psi \in C_{Z,N}\}} \langle \Psi | H_{\text{Lucas}} - H_{\text{rest}} | \Psi \rangle$$

17. Experimental and computational program

The hierarchy produces a direct verification sequence. Each stage must be completed before the next is allowed to introduce a new coefficient.

Stage	Operation	Pass condition
1. Geometry	Reconstruct B, M, R, W, G _{phase} , 27 nodes, 54 links, and all plane families.	All printed invariants and closure identities reproduce within declared numerical precision.
2. Local spectroscopy	Measure or compute transition lines among selected Cell3 node states.	Observed ratios follow the W-derived transition ledger after one fixed energy calibration.
3. Emily coupling	Excite the 27-node local graph and resolve the seven Emily levels.	Multiplicities 1,3,6,7,6,3,1 and one universal J _E .

Stage	Operation	Pass condition
4. A2 phase	Drive the two independent same-sheet phases.	Cold state, conjugate hot states, odd gradient, and even Hessian are reproduced.
5. Lucas cavity	Construct a coherent 3 x 3 x 3 device with controlled interfaces.	Twenty-seven phase sectors, 343-node intersections, 19 sheets, and predicted mode ratios.
6. Few-body nuclear fit	Encode d, triton, helion, and alpha using fixed p=100 and n=111 channels.	One universal closure operator reproduces signs, hierarchy, pairing, mirror shifts, and binding scale.
7. Shell progression	Extend occupation through the signed direction families and recursion.	Closures at 2,8,20,28 and later 50,82,126 emerge without insertion as labels.
8. Full mass surface	Predict separation energies and isotope chains from the frozen operator.	Residual structure is lower-dimensional and follows closure variables rather than nuclide-specific corrections.

The strongest falsification conditions are equally direct. The hierarchy fails if one universal dimensionalization cannot organize local transition energies, if the Lucas phase sectors do not form the exact Z_3^3 basis, if the Emily degeneracies fail under controlled equal-coupling conditions, if alpha closure cannot be distinguished from two independent deuteron bonds, or if the nuclear operator requires independent parameters for each nucleus.

18. Integrated energy hierarchy

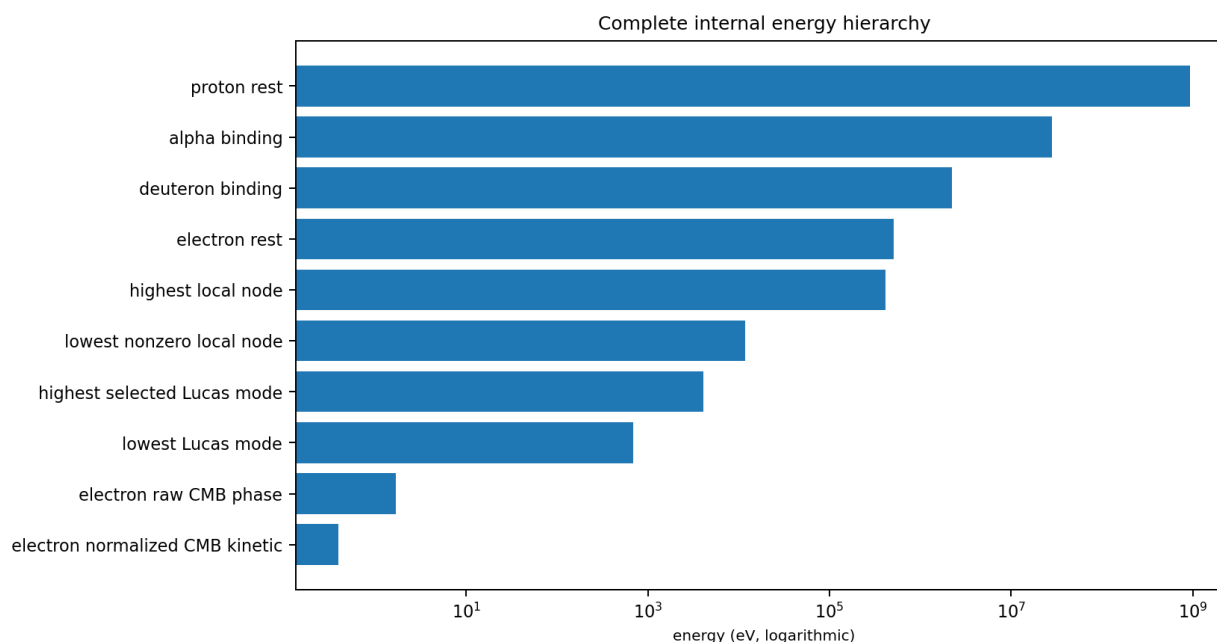


Figure 14. Complete internal energy hierarchy from the electron Earth-CMB rate state through local nodes, Lucas collective modes, nuclear binding, and nucleon rest energy.

Layer	Representative energy	Origin
electron normalized CMB kinetic	0.388787 eV	self-indexed traversal
electron raw CMB phase	1.666242 eV	anisotropic phase scalar
Lucas periodic modes	682.998 to 4098.036 eV	3B reciprocal spectrum
local node energy	11.796 to 416.806 keV	kappa_0 U_L
electron rest	0.510998951 MeV	electron realization
reciprocal radial bases	0.510999 to 1.475342 MeV	E_G s_k
deuteron binding	2.224566363 MeV	minimal pn closure
alpha binding	28.295610888 MeV	complete 2p+2n closure
proton rest	938.272089 MeV	1000 x Mass-Bridge radial

The hierarchy shows that the electron rest anchor supplies dimensional currency; Cell3 supplies local position, phase, and transition structure; the Earth-CMB vector supplies a rate-state bias; the full Lucas Group supplies collective closure; and nuclear mass defect is the energy released when free reciprocal rest states reorganize into a lower composite closure state.

19. Formula ledger

Object	Formula
Position	$x=(1/2)Bq$
Direct metric	$M=B^T B$
Reciprocal metric	$R=M^{-1}$
Phase metric	$G_{\text{phase}}=4B^T W B^{-1}$
Mass Bridge	$1000 A_{bc}/A_{ab}=\eta; 100 A_{bc}/A_{ac}=\alpha^{-1}$
Electron realization	$E_G=m_e c^2; L_G= a+b+c $
Rate state	$\dot{q}=2B^{-1} v$
Raw phase energy	$\Phi=\dot{q}^T W \dot{q}=v^T G_{\text{phase}} v$
Self-indexed kinetic energy	$E/m=[1/(2C(u))]v^T G_{\text{phase}} v=v^2/2$
Node energy index	$U_L=(1/2)L^T W L$
Canonical dimensionalization	$\kappa_0=E_G/L_G^2$
Transition increment	$\Delta_i+U=(WL)_i+W_{ii}/2$
Discrete gradient	$g_D=-2 \kappa_W B^T W L$
Reciprocal radial	$s_k=\sqrt{(k^T R k)/R_{cc}}$
Rest-energy state	$E_{\text{rest}}=E_G(p/d)s_k$
Emily spectrum	$\lambda=(m_a+m_b+m_c)\sqrt{2}$
A2 compact energy	$U_K=\sum w_{ij}[1-\cos \psi_{ij}]$
Quotient-fiber factor	$\chi_{QF}=1/(1-\rho_{QF}^2)$
Lucas mode	$E_N=[E_G L_G/(3S_{\text{CMB}})]\sqrt{N^T R N}$
Mixed CMB term	$\epsilon_{\text{cross}}=\kappa_F \dot{q}^T W \dot{q}_E$
Binding energy	$B=E_{\text{free}}-\min E_{\text{composite}}$
Incidence candidate	$B/E_G=I_{\text{lock}}+\text{trace}(W)+\delta_{\text{phase}}$

Appendix A. Authoritative matrices and invariants

Matrix	Rows
B (nm)	[3.275012420566392e-01, 3.533077584239210e-02, -1.012931763000165e-02] [-5.489700788499677e-02, 2.073198504382919e-01, -3.897877023705170e-01] [-3.748750289576525e-02, -5.474010853912257e-01, -6.897344083675747e-01]
M (nm ²)	[1.116760578967268e-01, 2.071033328072946e-02, 4.393723509648568e-02] [2.071033328072946e-02, 3.438777323948730e-01, 2.963927589633223e-01] [4.393723509648568e-02, 2.963927589633223e-01, 6.277706100811045e-01]
R (nm ⁻²)	[9.208025529401748e+00, 1.534802428262973e-03, -6.451880356960570e-01] [1.534802428263103e-03, 4.903398312338273e+00, -2.315175584402374e+00] [-6.451880356960570e-01, -2.315175584402374e+00, 2.731171274695146e+00]
W (nm ²)	[8.335330263345708e-02, 1.375005766035881e-01, 1.025592112219582e-01] [1.375005766035881e-01, 6.924000428485476e-01, 5.508009044534695e-01] [1.025592112219582e-01, 5.508009044534695e-01, 5.877476523533913e-01]
G_phase	[2.976410638029592e+00, 1.243150779538656e+00, -2.080360035502319e+00] [1.243150779538656e+00, 2.592228663506714e+00, -2.317883993755083e+00] [-2.080360035502319e+00, -2.317883993755082e+00, 6.773570913318284e+00]

Invariant	Value
det(B)	-1.179148919128169e-01 nm ³
det(B)	1.179148919128169e-01 nm ³
trace(W)	1.363500997835396 nm ²
W eigenvalues	0.050269586792171, 0.094205549207267, 1.219025861835957
W condition ratio	24.249768888607637
G_phase eigenvalues	1.429328753501001, 2.112545026528079, 8.800336434825510
U_hot	1147.143255372121985
chi_QF	2.191548594807890

Appendix B. Complete 27-node Cell3 ledger

Address	Class	lambda	chi	x (nm)	y (nm)	z (nm)	U (nm^2)	E^(0) (keV)
-1,-1,-1	corner	-3	-1	-0.176351350	0.118682430	0.637311498	1.472611191	416.805509
-1,-1,+0	edge center	-2	1	-0.181416009	-0.076211421	0.292444294	0.525377249	148.701934
-1,+0,-1	edge center	-2	1	-0.158685962	0.222342355	0.363610956	0.438109689	124.001864
+0,-1,-1	edge center	-2	1	-0.012600729	0.091233926	0.618567747	1.190874752	337.063280
-1,-1,+1	corner	-1	-1	-0.186480668	-0.271105272	-0.052422910	0.165890960	46.953511
-1,+0,+0	face center	-1	-1	-0.163750621	0.027448504	0.018743751	0.041676651	11.796093
-1,+1,-1	corner	-1	-1	-0.141020574	0.326002280	0.089910413	0.096008229	27.174015
+0,-1,+0	face center	-1	-1	-0.017665388	-0.103659925	0.273700543	0.346200021	97.987899
+0,+0,-1	face center	-1	-1	0.005064659	0.194893851	0.344867204	0.293873826	83.177576
+1,-1,-1	corner	-1	-1	0.151149892	0.063785422	0.599823995	0.992491616	280.913235
-1,+0,+1	edge center	0	1	-0.168815280	-0.167445347	-0.326123453	0.232991266	65.945474
-1,+1,+0	edge center	0	1	-0.146085233	0.131108429	-0.254956791	0.250376096	70.866049
+0,-1,+1	edge center	0	1	-0.022730047	-0.298553776	-0.071166661	0.089272943	25.267671
+0,+0,+0	body center	0	1	0.000000000	0.000000000	0.000000000	0.000000000	0.000000
+0,+1,-1	edge center	0	1	0.022730047	0.298553776	0.071166661	0.089272943	25.267671
+1,-1,+0	edge center	0	1	0.146085233	-0.131108429	0.254956791	0.250376096	70.866049
+1,+0,-1	edge center	0	1	0.168815280	0.167445347	0.326123453	0.232991266	65.945474
-1,+1,+1	corner	1	-1	-0.151149892	-0.063785422	-0.599823995	0.992491616	280.913235
+0,+0,+1	face center	1	-1	-0.005064659	-0.194893851	-0.344867204	0.293873826	83.177576
+0,+1,+0	face center	1	-1	0.017665388	0.103659925	-0.273700543	0.346200021	97.987899
+1,-1,+1	corner	1	-1	0.141020574	-0.326002280	-0.089910413	0.096008229	27.174015
+1,+0,+0	face center	1	-1	0.163750621	-0.027448504	-0.018743751	0.041676651	11.796093
+1,+1,-1	corner	1	-1	0.186480668	0.271105272	0.052422910	0.165890960	46.953511
+0,+1,+1	edge center	2	1	0.012600729	-0.091233926	-0.618567747	1.190874752	337.063280
+1,+0,+1	edge center	2	1	0.158685962	-0.222342355	-0.363610956	0.438109689	124.001864
+1,+1,+0	edge center	2	1	0.181416009	0.076211421	-0.292444294	0.525377249	148.701934
+1,+1,+1	corner	3	-1	0.176351350	-0.118682430	-0.637311498	1.472611191	416.805509

Appendix C. Complete 54-link transition ledger

From	To	Ch.	lambda_i	lambda_f	E_i (keV)	E_f (keV)	Delta E (keV)	Delta (keV)
-1,-1,-1	+0,-1,-1	a	-3	-2	416.805509	337.063280	-79.742230	79.742230
-1,-1,-1	-1,+0,-1	b	-3	-2	416.805509	124.001864	-292.803646	292.803646
-1,-1,-1	-1,-1,+0	c	-3	-2	416.805509	148.701934	-268.103575	268.103575
-1,-1,+0	+0,-1,+0	a	-2	-1	148.701934	97.987899	-50.714035	50.714035
-1,-1,+0	-1,+0,+0	b	-2	-1	148.701934	11.796093	-136.905841	136.905841
-1,-1,+0	-1,-1,+1	c	-2	-1	148.701934	46.953511	-101.748423	101.748423
-1,-1,+1	+0,-1,+1	a	-1	0	46.953511	25.267671	-21.685841	21.685841
-1,-1,+1	-1,+0,+1	b	-1	0	46.953511	65.945474	+18.991963	18.991963
-1,+0,-1	+0,+0,-1	a	-2	-1	124.001864	83.177576	-40.824287	40.824287
-1,+0,-1	-1,+1,-1	b	-2	-1	124.001864	27.174015	-96.827848	96.827848
-1,+0,-1	-1,+0,+0	c	-2	-1	124.001864	11.796093	-112.205771	112.205771
-1,+0,+0	+0,+0,+0	a	-1	0	11.796093	0.000000	-11.796093	11.796093
-1,+0,+0	-1,+1,+0	b	-1	0	11.796093	70.866049	+59.069956	59.069956
-1,+0,+0	-1,+0,+1	c	-1	0	11.796093	65.945474	+54.149382	54.149382
-1,+0,+1	+0,+0,+1	a	0	1	65.945474	83.177576	+17.232102	17.232102
-1,+0,+1	-1,+1,+1	b	0	1	65.945474	280.913235	+214.967761	214.967761
-1,+1,-1	+0,+1,-1	a	-1	0	27.174015	25.267671	-1.906345	1.906345
-1,+1,-1	-1,+1,+0	c	-1	0	27.174015	70.866049	+43.692033	43.692033
-1,+1,+0	+0,+1,+0	a	0	1	70.866049	97.987899	+27.121850	27.121850
-1,+1,+0	-1,+1,+1	c	0	1	70.866049	280.913235	+210.047186	210.047186
-1,+1,+1	+0,+1,+1	a	1	2	280.913235	337.063280	+56.150045	56.150045
+0,-1,-1	+1,-1,-1	a	-2	-1	337.063280	280.913235	-56.150045	56.150045
+0,-1,-1	+0,+0,-1	b	-2	-1	337.063280	83.177576	-253.885703	253.885703
+0,-1,-1	+0,-1,+0	c	-2	-1	337.063280	97.987899	-239.075381	239.075381
+0,-1,+0	+1,-1,+0	a	-1	0	97.987899	70.866049	-27.121850	27.121850
+0,-1,+0	+0,+0,+0	b	-1	0	97.987899	0.000000	-97.987899	97.987899
+0,-1,+0	+0,-1,+1	c	-1	0	97.987899	25.267671	-72.720228	72.720228
+0,-1,+1	+1,-1,+1	a	0	1	25.267671	27.174015	+1.906345	1.906345
+0,-1,+1	+0,+0,+1	b	0	1	25.267671	83.177576	+57.909906	57.909906
+0,+0,-1	+1,+0,-1	a	-1	0	83.177576	65.945474	-17.232102	17.232102
+0,+0,-1	+0,+1,-1	b	-1	0	83.177576	25.267671	-57.909906	57.909906
+0,+0,-1	+0,+0,+0	c	-1	0	83.177576	0.000000	-83.177576	83.177576
+0,+0,+0	+1,+0,+0	a	0	1	0.000000	11.796093	+11.796093	11.796093
+0,+0,+0	+0,+1,+0	b	0	1	0.000000	97.987899	+97.987899	97.987899
+0,+0,+0	+0,+0,+1	c	0	1	0.000000	83.177576	+83.177576	83.177576
+0,+0,+1	+1,+0,+1	a	1	2	83.177576	124.001864	+40.824287	40.824287
+0,+0,+1	+0,+1,+1	b	1	2	83.177576	337.063280	+253.885703	253.885703
+0,+1,-1	+1,+1,-1	a	0	1	25.267671	46.953511	+21.685841	21.685841
+0,+1,-1	+0,+1,+0	c	0	1	25.267671	97.987899	+72.720228	72.720228
+0,+1,+0	+1,+1,+0	a	1	2	97.987899	148.701934	+50.714035	50.714035
+0,+1,+0	+0,+1,+1	c	1	2	97.987899	337.063280	+239.075381	239.075381
+0,+1,+1	+1,+1,+1	a	2	3	337.063280	416.805509	+79.742230	79.742230
+1,-1,-1	+1,+0,-1	b	-1	0	280.913235	65.945474	-214.967761	214.967761
+1,-1,-1	+1,-1,+0	c	-1	0	280.913235	70.866049	-210.047186	210.047186
+1,-1,+0	+1,+0,+0	b	0	1	70.866049	11.796093	-59.069956	59.069956
+1,-1,+0	+1,-1,+1	c	0	1	70.866049	27.174015	-43.692033	43.692033
+1,-1,+1	+1,+0,+1	b	1	2	27.174015	124.001864	+96.827848	96.827848
+1,+0,-1	+1,+1,-1	b	0	1	65.945474	46.953511	-18.991963	18.991963

From	To	Ch.	lambda_i	lambda_f	E_i (keV)	E_f (keV)	Delta E (keV)	Delta (keV)
+1,+0,-1	+1,+0,+0	c	0	1	65.945474	11.796093	-54.149382	54.149382
+1,+0,+0	+1,+1,+0	b	1	2	11.796093	148.701934	+136.905841	136.905841
+1,+0,+0	+1,+0,+1	c	1	2	11.796093	124.001864	+112.205771	112.205771
+1,+0,+1	+1,+1,+1	b	2	3	124.001864	416.805509	+292.803646	292.803646
+1,+1,-1	+1,+1,+0	c	1	2	46.953511	148.701934	+101.748423	101.748423
+1,+1,+0	+1,+1,+1	c	2	3	148.701934	416.805509	+268.103575	268.103575

Appendix D. Complete 13-channel reciprocal radial ledger

Radial	Description	q (nm ⁻²)	s_k	E_G s_k (MeV)	E_node ^Λ (0) (keV)
001	primitive c-axis	2.731171275	1.000000000000	0.510998951	83.177576
011	bc face diagonal (+)	3.004218418	1.048796632266	0.535933979	337.063280
010	primitive b-axis	4.903398312	1.339905341993	0.684690224	97.987899
100	primitive a-axis / Mass-Bridge	9.208025529	1.836152673426	0.938272089	11.796093
101	ac face diagonal (+)	10.648820733	1.974587075075	1.009011923	124.001864
111	body diagonal (+++)	10.924937481	2.000023101888	1.022009706	416.805509
01-1	bc face diagonal (-)	12.264920756	2.119131490765	1.082873968	25.267671
10-1	ac face diagonal (-)	13.229572875	2.200890538669	1.124652756	65.945474
1-1-1	body diagonal (+--)	13.499550414	2.223234046131	1.136070265	280.913235
1-10	ab face diagonal (-)	14.108354237	2.272813025845	1.161405071	70.866049
110	ab face diagonal (+)	14.114493447	2.273307476082	1.161657735	148.701934
1-11	body diagonal (+-+)	20.179500609	2.718195812855	1.388995208	27.174015
11-1	body diagonal (++-)	22.766391961	2.887171860755	1.475341791	46.953511

Appendix E. Complete 19-sheet Lucas Group ledger

lambda	Population	Interior	Boundary	chi	Offset (nm)	Total U	Mean U
-9	1	0	1	-1	-2.327276340524	13.253500721	13.253500721
-8	3	0	3	1	-2.068690080466	31.606585514	10.535528505
-7	6	0	6	-1	-1.810103820407	49.059699422	8.176616570
-6	10	1	9	1	-1.551517560349	61.767649176	6.176764918
-5	15	3	12	-1	-1.292931300291	68.039603198	4.535973547
-4	21	6	15	1	-1.034345040233	68.339091601	3.254242457
-3	28	10	18	-1	-0.775758780175	65.284006187	2.331571650
-2	33	15	18	1	-0.517172520116	52.329993869	1.585757390
-1	36	18	18	-1	-0.258586260058	40.248863098	1.118023975
0	37	19	18	1	0.000000000000	35.503698944	0.959559431
1	36	18	18	-1	0.258586260058	40.248863098	1.118023975
2	33	15	18	1	0.517172520116	52.329993869	1.585757390
3	28	10	18	-1	0.775758780175	65.284006187	2.331571650
4	21	6	15	1	1.034345040233	68.339091601	3.254242457
5	15	3	12	-1	1.292931300291	68.039603198	4.535973547
6	10	1	9	1	1.551517560349	61.767649176	6.176764918
7	6	0	6	-1	1.810103820407	49.059699422	8.176616570
8	3	0	3	1	2.068690080466	31.606585514	10.535528505
9	1	0	1	-1	2.327276340524	13.253500721	13.253500721

Appendix F. Data provenance and reproducibility

Source	Role
K-Field Mass-Bridge Is Now a Locked Projective Ratio System, epoch 1785176592	Exact eta-alpha face lock and forced Cell3 shape.
K-Field Linear Inertia Derivation, epoch 1784989727	W, G_phase, Earth-CMB rate state, raw and normalized phase energy.
K-Field Devon / Lucas / Emily / Jack, epoch 1784245224	Seven sheets, Emily spectrum, Jack perturbation grammar.
K-Field Lucas Group Standing-Wave Cavity Analysis, epoch 1784062416	Physical ICRS basis, 343-node group, 19 sheets, 27 phase sectors, periodic modes.
K-Field Invariants and Hidden Pathways, epoch 1783951701	A2 compact energy, quotient-fiber coupling, weighted branch Laplacian.
K-Field Mass Harmonics, epoch 1784575932	Thirteen reciprocal radial channels and particle rest-energy ladder.
Recursive Scale-Transition Nuclear Binding Subreport	Closure-lock energy requirements and free-to-composite mass-defect interpretation.
NIST, 2022 CODATA Recommended Values of the Fundamental Physical Constants	Electron, proton, neutron, deuteron, and alpha realization anchors.

Field	Record
PDF engine	Python ReportLab
Permitted PDF fonts	Helvetica, Helvetica-Bold, Times-Roman, Courier
Figure engine	Python Matplotlib, raster images generated from numerical ledgers
Page size	US Letter
Margins	0.68 in left/right/top, 0.72 in bottom
Table contract	All text cells are ReportLab Paragraph objects; explicit colWidths; repeatRows=1
Generated UTC	2026-07-27T22:21:40Z
Unix epoch	1785190900
Machine-readable companion	K-Field_internal_energy_hierarchy_1785190900.json
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.

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- [8] Recursive Scale-Transition Nuclear Binding in the Axis-Plane Closure Framework. Krytronx archival subreport.